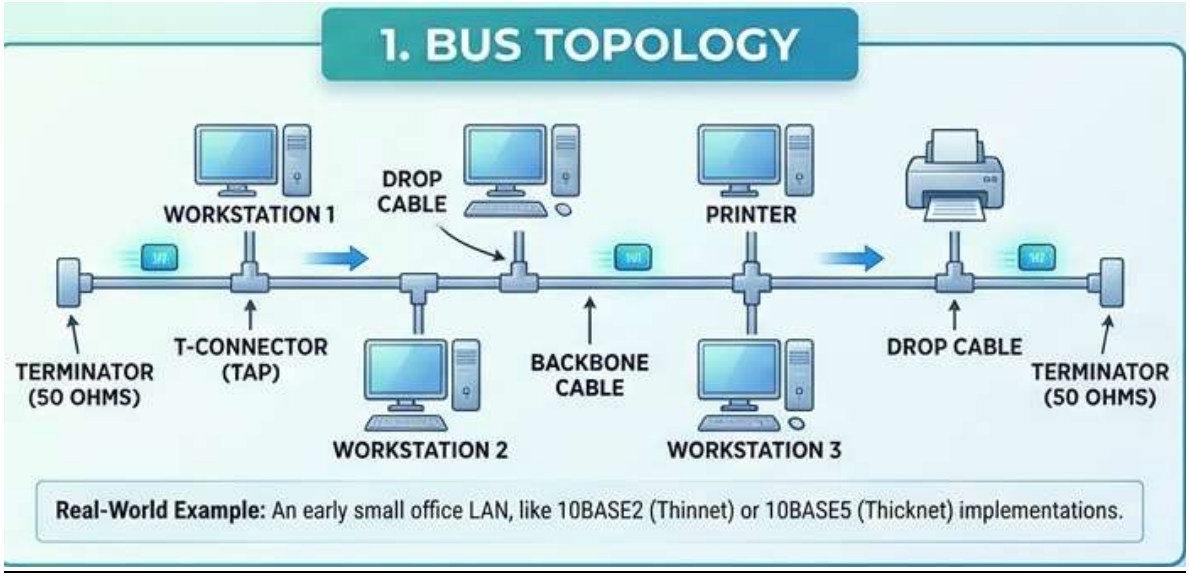
1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

ANS:-



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2. In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

ANS:-

Creating a Star Topology in Cisco Packet Tracer (Theory)

A star topology is a network arrangement in which all computers are connected to a central networking device, usually a switch. In Cisco Packet Tracer, a star topology can be created by connecting four PCs to one switch using Ethernet cables.

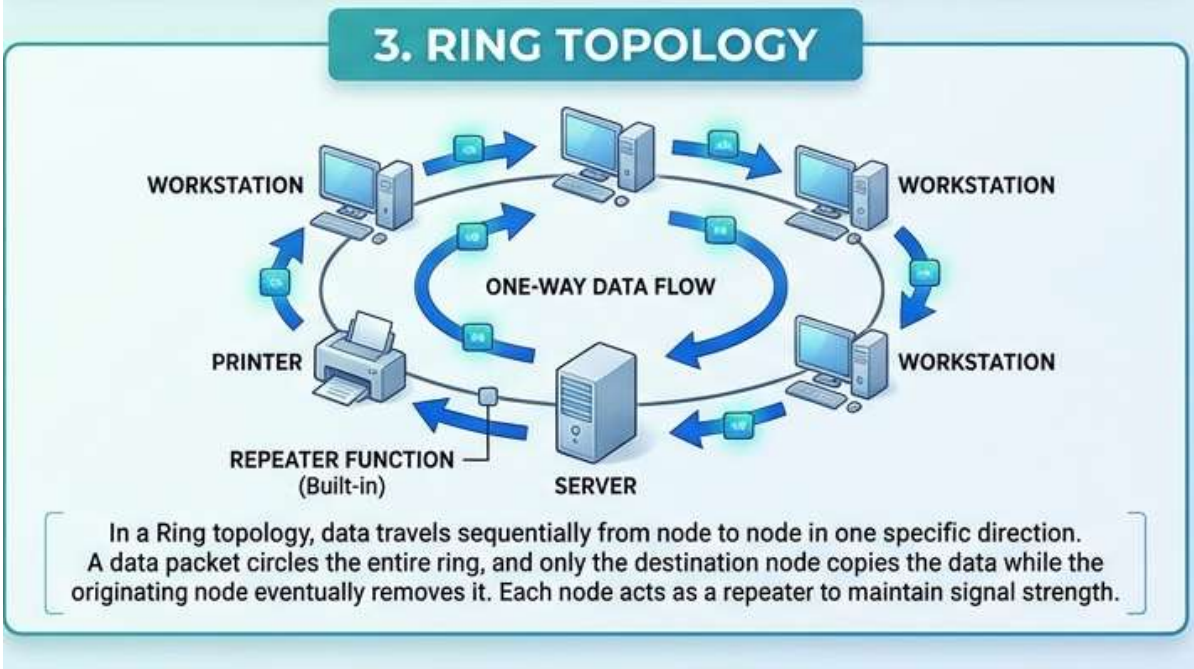
Procedure:

1. Open Cisco Packet Tracer and place one switch and four PCs in the workspace.
2. Rename the PCs as WhatsApp, Zomato, Spotify, and Flipkart.
3. Connect each PC to a separate port on the switch using Copper Straight-Through cables.
4. Wait for the connection indicators to turn green.
5. Arrange the devices so that the switch is in the center and the four PCs are around it.
6. Save the topology and take a screenshot for the assignment.

Result: A star topology consists of four PCs connected to a central switch. The switch forwards data between devices, and if one PC or its cable fails, the other PCs can continue communicating. However, if the central switch fails, communication between the connected devices is disrupted.

3. Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup. Hint: Use arrows to show the direction of data flow.

ANS:-



4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

ANS:-

Comparison Between Mesh and Star Topologies

Feature	**Star Topology**	**Mesh Topology**
Advantages	1. Easy to install and manage. 2. Failure of one device or cable usually does not affect other devices.	1. Provides multiple paths for data transmission. 2. Offers high reliability and fault tolerance.
Disadvantages	1. Failure of the central switch can stop the entire network. 2. Requires more cabling than some simpler topologies.	1. Requires more cables and connections. 2. Expensive and complex to install and maintain.

Conclusion: Star topology is suitable for small LANs and computer labs because it is simple and cost-effective. Mesh topology is useful when high reliability and alternative communication paths are important.